

Industrial Controls



Industrial Controls

Industrial control devices range from simple knife switches to more complex solid state sensors. The type of control device selected depends upon the specific application. There are three distinct ways to control an application:

- Manually Operated Switches
- Mechanically Operated Switches
- Electromechanical Switches



Industrial Controls

- Manually Operated Switches

Manually operated switches are triggered with the use of human power. There are five types of manually operated switches:

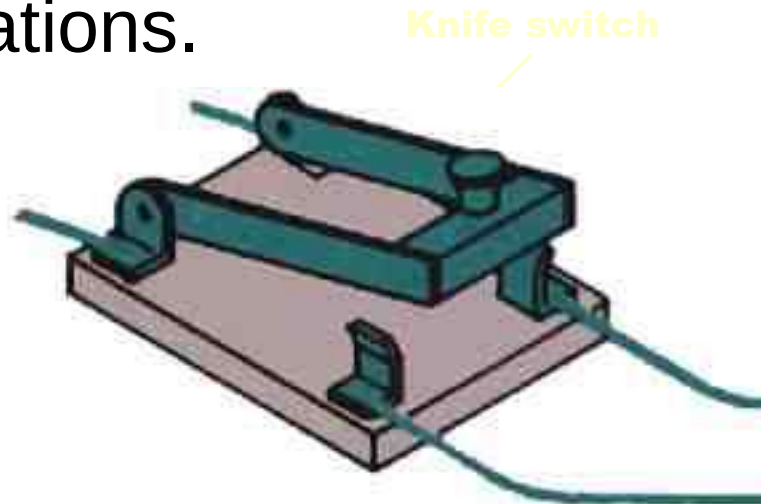
- Knife
- Toggle
- Push Button
- Rotary-Selector
- Manual Starter



Knife Switch

The knife switch is the oldest and simplest type of switch consisting of a blade and jaw mechanism.

Knife switches are primarily used in large electrical distribution situations.



Toggle Switch

Toggle switches replaced knife switches for safety and speed of operation. Toggle switches use a lever to close or open contacts.



Toggle switch



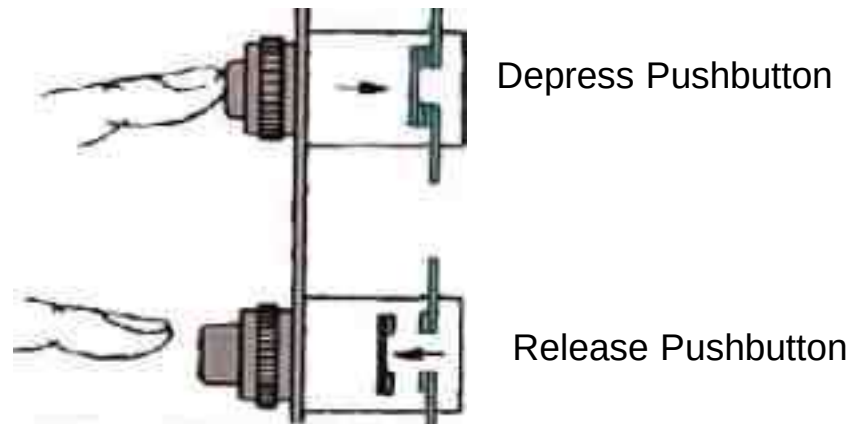
Pushbutton Switch

A pushbutton switch is a two-position switch used to select a load either on or off.

The pushbutton may consist of one or more contact blocks; denoting the switch as normally open (no) or normally closed (nc), an operator of some type, a legend plate to denote function and mounting rings or washers to adjust and hold the device in an enclosure.



Pushbutton Switches



Rotary-Selector Switch

Rotary-selector switches are constructed in the same fashion as a pushbutton. A rotary-selector switch requires the operator to rotate a knob, lever, or key to open and close contacts. Additionally, rotary-selector switches are used to select or determine one of several different circuit possibilities from a two-position on/off switch to a four-position key switch.



Coin

Knob

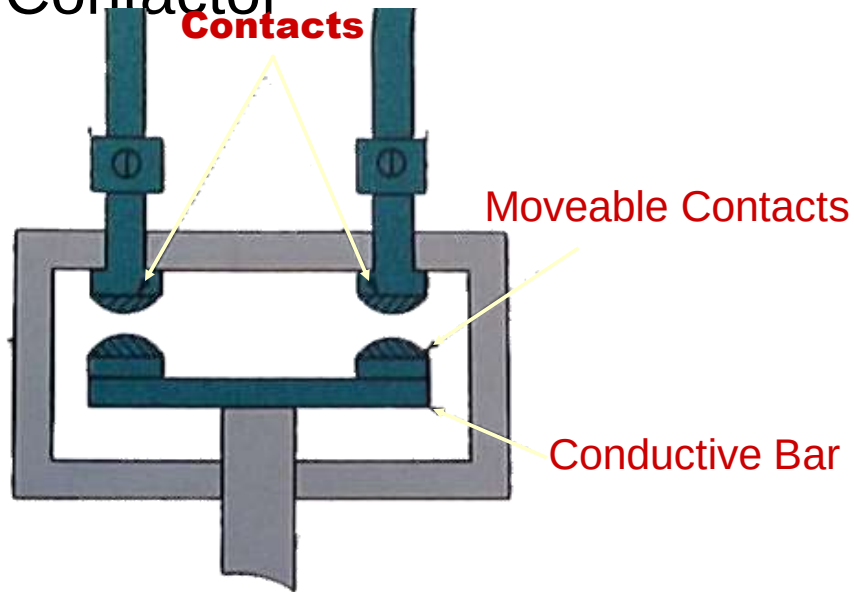
Key

Rotary-Selector Switch

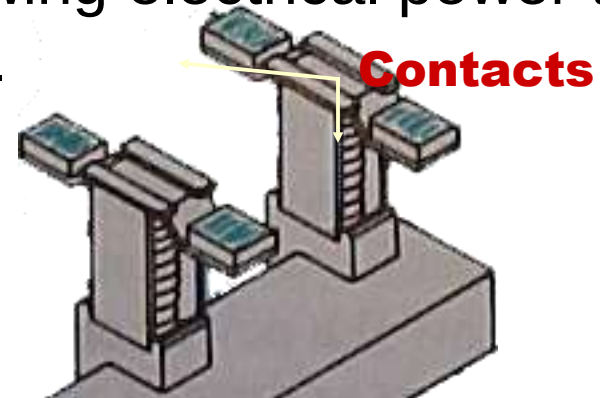


Manual Contactor

Manual
Contactor

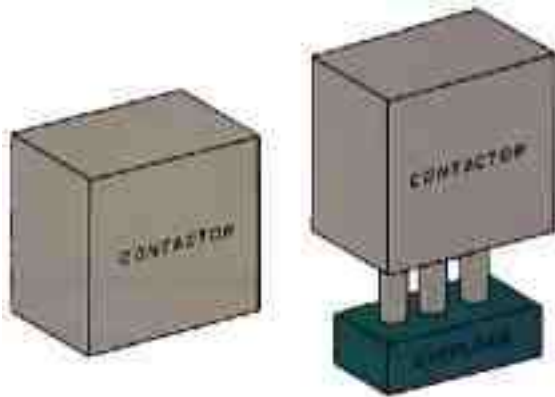


A manual contactor is a device for any electrical circuit that repeatedly establishes and interrupts an electrical power circuit. When a switch is engaged, contacts touch allowing electrical power to flow.



Manual Starter Switch

Manual Starter



Manual Starter switches are contactors with an additional overload protection device used only for motors. The overload protection device is required to protect the motor from destroying itself under an overloaded situation.



Industrial Controls

- **Mechanically Operated Switches**

Mechanically operated switches are triggered with the use of some other moving part. There are four types of mechanically operated switches:

- Limit
- Mercury
- Snap Acting
- Motor Protection



Limit Switch

Limit switches are used to convert mechanical motion into an electrical signal. They accomplish this conversion by using some type of lever to force open or closed a set of contacts within the limit switch.

Typical applications of limit switches include conveyors, elevator control, counting, positioning, detecting, sequencing, and monitoring.

Limit Switches



Mercury Switch

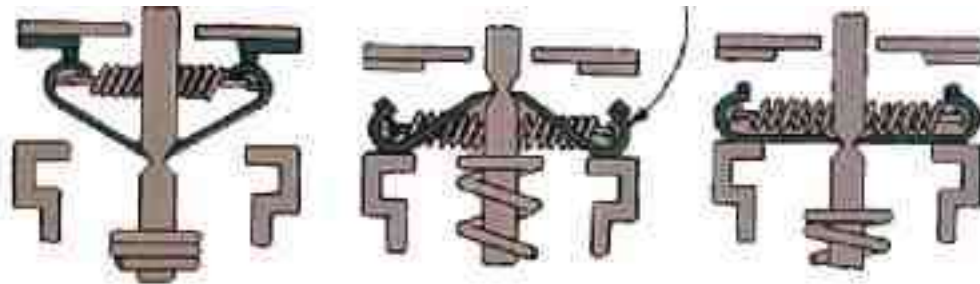
Mercury switches utilize the electrical properties of mercury to complete a circuit. Within a glass tube, a pool of mercury is able to move around allowing it to make contact with different contacts for different reasons. Usually a mercury switch is used in low voltage situations such as a household thermostat.



Snap Acting Switch

A snap acting switch is used in applications that require a set of contacts that will snap open or closed to eliminate arcing and burning.

Bimetallic Strip

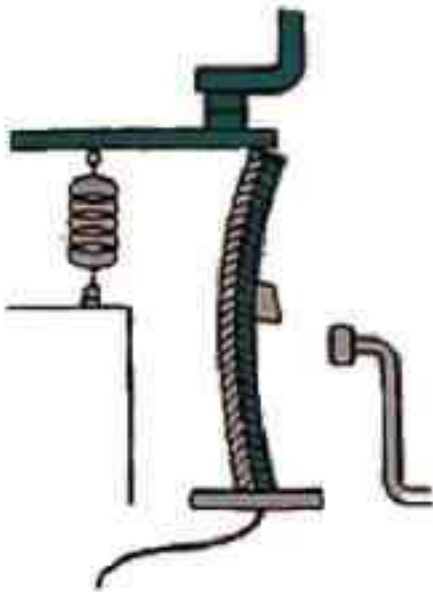


As the bimetallic strip heats or cools, pressure is exerted on the spring, eventually causing enough force to instantly snap the plunger up or down

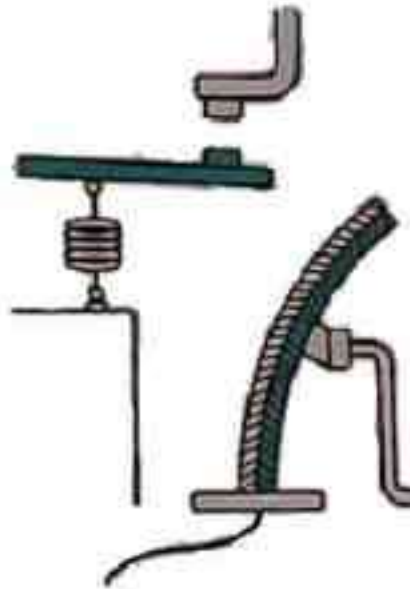


Motor Protection Switch

Before Overload



After Overload



Motor protection switches are located on or in a motor to protect the motor from overloading situations. They sense either the amount of heat generated or the amount of current consumed by the motor. Prior to motor damage, they directly or indirectly disconnect the motor from the power circuit.



Industrial Controls

- Electromechanical Switches

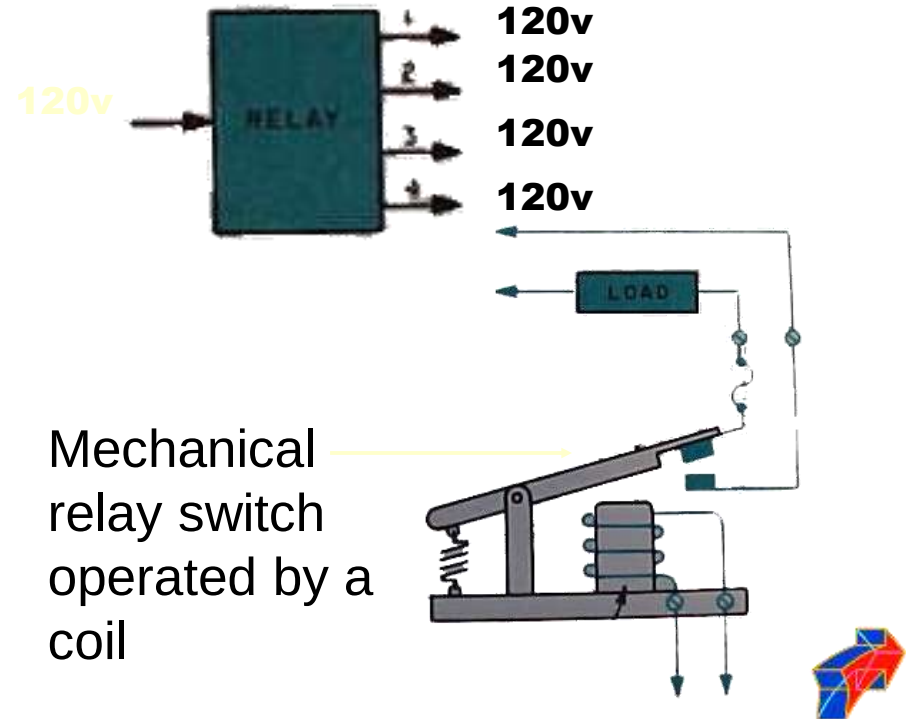
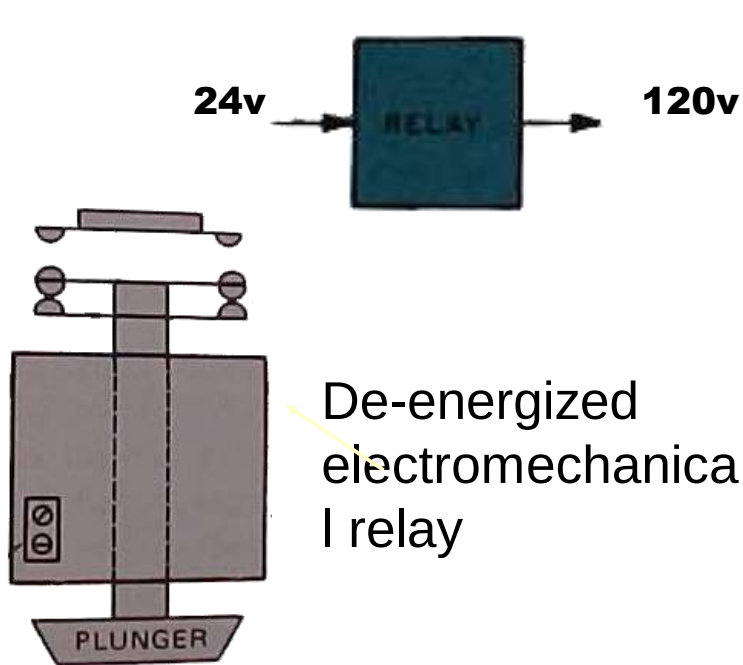
Electromechanical operated switches are triggered with the use electricity. There are three types of electromechanically operated switches:

- Relays
- Solenoids
- Semi-conductive



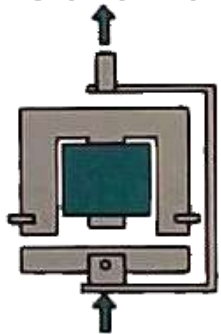
Relays

Relays are electromechanical devices that either use a small input voltage (24v) to control a larger output voltage (230/460v) or use an input voltage to control two or more output voltages.

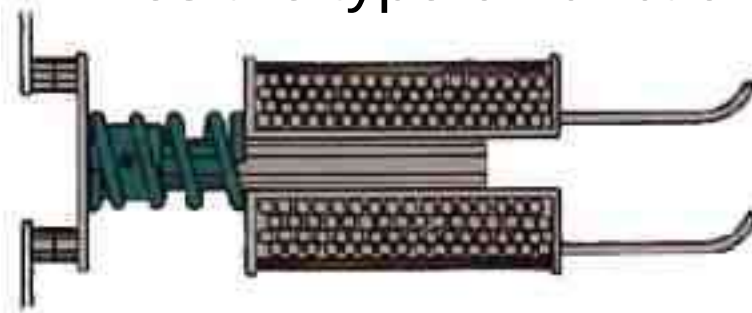


Solenoid Switches

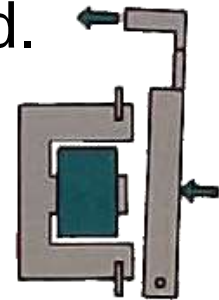
Solenoid switches use simple electromagnetic attraction and repulsion to create some type of useful function. The design of the solenoid determines the type of function created.



Vertical Action



N.C. contacts open when energized



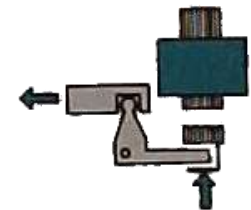
Clapper



Horizontal Action



N.O. contacts close when energized



Bell Crank



Semi-conductive Switch

Semi-conductive switches use semiconductor technology to control low voltage electricity. Examples include rectifiers, silicon controlled rectifiers, diodes, thermistors, photovoltaic cells, photoconductive cells, resistors, transistors, capacitors and integrated circuits.

